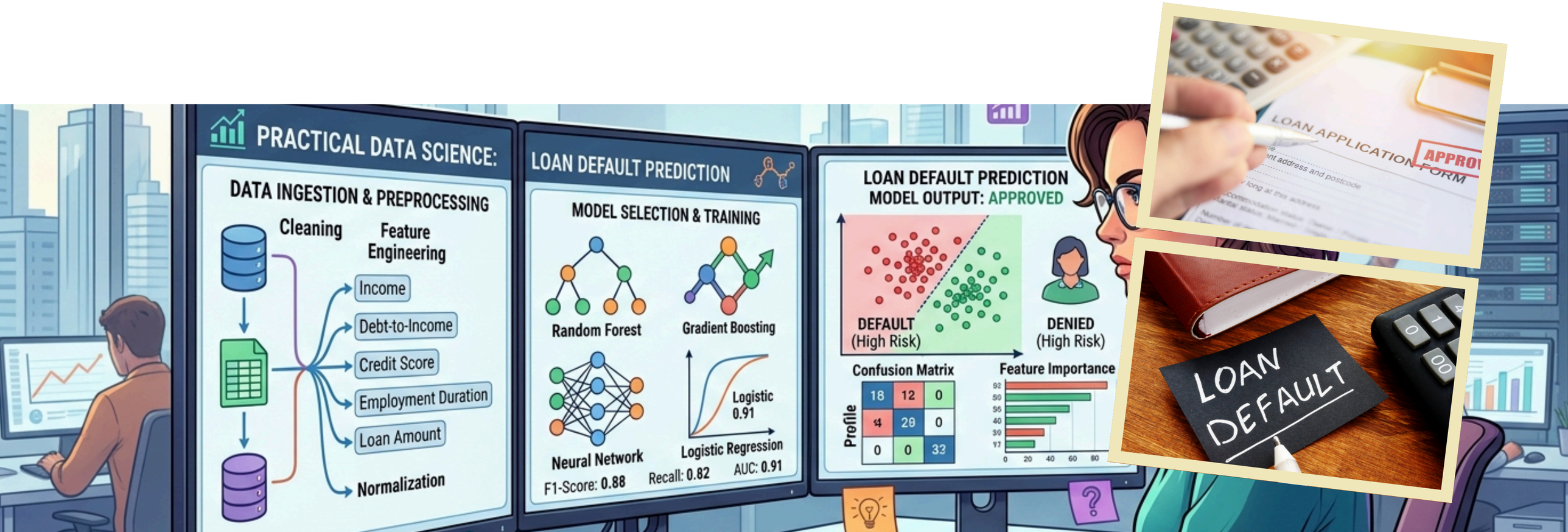
Capstone Project - Practical Data Science

Loan Default Prediction

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PROJECT INTRODUCTION

- Home loans are a primary source of revenue for retail banks → It is extremely important that the loan process approval is as much accurate as possible.

- Defaulted loans are a big problem for banks → A lot of defaulted loans will lead to significant financial losses if not properly identified beforehand.

- The approval process is time consuming and multifaceted → It is difficult to not face human errors and biases through the whole process.

PROBLEM STATEMENT & OBJECTIVES

We want to reduce at minimum the human bias component by building a reliable predictive model based on loan history → we need to be able to accurately classify prospective applicants who are likely to default, based on specific parameters, in order to *automate/assist* the loan approval process.

This is a ***supervised binary classification*** problem. We are using historical tabular data of past applicants to predict a discrete, binary target variable (BAD, where 1 = Client defaulted on loan, 0 = loan repaid).

Objectives:

- Find the most important applicant characteristics (e.g., credit history, job type, debt-to-income ratio) that are the *main drivers of loan default*.
- Build a predictive model that best balances the trade-off between catching risky loans (*Recall*) and accidentally rejecting good customers (*Precision*).
- Also, determine which machine learning algorithm provides the best balance of *accuracy* and *interpretability*.

DATA OVERVIEW

- The dataset collects performance information about **5960 recent home equity loans**.

- There are **12 input attributes** registered for every record, plus the target variable BAD. 10 of the 12 input attributes are numerical, with only 2 being categorical.

- The **target variable BAD** is a dummy representing: 1 = Client defaulted on loan, 0 = loan repaid.

- Based on the distribution of BAD, the **data is unbalanced**: only 20% of the data is labelled as “bad” loans.

EXPLORATORY DATA ANALYSIS

- Most variables contain missing values, especially DEBTINC, making data imputation (data filling) an important step.

- The majority of loans are non-defaults (BAD = 0), indicating class imbalance in the dataset.

- Variables like LOAN, YOJ, DELINQ, and DEROG are right-skewed, with most values concentrated at lower ranges and a few extreme observations.

- Extreme values in variables like DEBTINC, CLAGE, VALUE, and MORTDUE suggest outliers or possible data inconsistencies.

- It is interesting to note that most applicants have zero delinquencies and derogatory reports, indicating that the risk is concentrated in a smaller subset of applicants.

DATA PRE-PROCESSING

- Outliers have been clipped, through the *IQR Method with Capping*. Any value higher (lower) than the upper (lower) whisker gets substituted with the upper (lower) whisker itself.

- Missing values have been filled with the *median for numerical columns*, and with the *mode for categorical variables*.

- In order to build the predictive models, the original dataset has been split in 2 parts: one part containing the observations related to the target variable, the other containing all the independent variables. Then, these 2 parts have been split again into train and test set, in a ratio of 70:30 for train and test set.

MODELING TECHNIQUES

LOGISTIC REGRESSION

- It’s the baseline model.
- It’s also the weakest model for the bank's needs, as it fails to identify the majority of defaults.

DECISION TREES

- The initial decision tree tends to overfit the training data
- By pruning it, the model improves and becomes much more effective in catching defaults.

RANDOM FOREST

- The initial random forest model has again the same overfitting problem for the training data
- Applying hyperparameter tuning allows us to get a reliable model.

BOOSTING CLASSIFIERS

- Overall, these classifiers showed higher precision.
- Looking in particular at the XGBoost, this model reached the highest scores in terms of accuracy and precision.

MODEL COMPARISON

Model	Accuracy	Precision	Recall
Logistic Regression	*80.43%*	70.59%	3.36%
Decision Tree	85.63%	65.63%	58.82%
Tuned Decision Tree	86.52%	64.08%	73.95%
Random Forest	89.93%	78.64%	68.07%
Tuned Random Forest	87.36%	66.09%	75.35%
AdaBoost	86.91%	65.80%	71.71%
Gradient Boosting	89.49%	79.86%	63.30%
XGBoost	90.49%	83.27%	65.55%
LightGBM	90.43%	82.29%	66.39%

FINAL MODEL PROPOSAL

- For a bank, missing a default is usually much more expensive than accidentally rejecting a good customer. Therefore, banks generally prefer to maximize Recall, even if it impacts a little bit on Precision and Accuracy.

- Based on the results obtained from the analysis, the **Tuned Random Forest** is the most robust solution, *catching 3 out of 4 actual defaults* while maintaining an overall **87% accuracy rate** on unseen data and a **precision of about 66%**.

- Overall, it's the model with the **highest Recall** (**75%**).

IMPLEMENTATION RECOMMENDATIONS

- Use the model as an assistant: even if the Tuned Random Forest has been selected as the optimal solution, do not exclusively rely on it. Take advantage of it in order to draw an accurate conclusion for more difficult cases.

- Prioritize data collection for Debt-to-Income ratio: the feature importance analysis showed that DEBTINC and its missing value flag are the most important predictors of default. The bank should implement stricter requirements for applicants to provide this data, in order to increase model reliability.

- Address class imbalance in future training: given that the dataset is unbalanced (only 20% are "bad" loans), future implementations should explore techniques like SMOTE (Synthetic Minority Over-sampling Technique) or cost-sensitive learning to further improve the model's ability to identify the minority class.

CONCLUSION

- **Model Selection**: after comparing several machine learning techniques the Tuned Random Forest has been selected as the optimal solution.

- **Model Performance**: the final model achieved a Recall of 75.35%, successfully catching 3 out of every 4 potential defaults, while maintaining an overall Accuracy of 87.36%.

- **Business Impact**: by prioritizing Recall, the bank can significantly reduce the high costs associated with defaulted loans, which are far more expensive than the opportunity cost of accidentally rejecting a good customer.

- **Final Insight**: the analysis confirms that credit history and especially debt-to-income ratios remain the most critical drivers of default. Implementing this automated, data-driven approach will lead to a more reliable, efficient, and profitable loan approval process.

Thank You!







APPENDIX

DATA DESCRIPTION

- *BAD*: 1 = Client defaulted on loan, 0 = loan repaid
- *LOAN*: Amount of loan approved.
- *MORTDUE*: Amount due on the existing mortgage.
- *VALUE*: Current value of the property.
- *REASON*: Reason for the loan request. (HomeImp = home improvement, DebtCon= debt consolidation which means taking out a new loan to pay off other liabilities and consumer debts)
- *JOB*: The type of job that loan applicant has such as manager, self, etc.
- *YOJ*: Years at present job.
- *DEROG*: Number of major derogatory reports (which indicates a serious delinquency or late payments).
- *DELINQ*: Number of delinquent credit lines (a line of credit becomes delinquent when a borrower does not make the minimum required payments 30 to 60 days past the day on which the payments were due).
- *CLAGE*: Age of the oldest credit line in months.
- *NINQ*: Number of recent credit inquiries.
- *CLNO*: Number of existing credit lines.
- *DEBTINC*: Debt-to-income ratio (all your monthly debt payments divided by your gross monthly income. This number is one way lenders measure your ability to manage the monthly payments to repay the money you plan to borrow.

SUMMARY STATISTICS

	count	mean	std	min	25%	50%	75%	max
LOAN	5960.0	18607.969799	11207.480417	1100.000000	11100.000000	16300.000000	23300.000000	89900.000000
MORTDUE	5442.0	73760.817200	44457.609458	2063.000000	46276.000000	65019.000000	91488.000000	399550.000000
VALUE	5848.0	101776.048741	57385.775334	8000.000000	66075.500000	89235.500000	119824.250000	855909.000000
YOJ	5445.0	8.922268	7.573982	0.000000	3.000000	7.000000	13.000000	41.000000
DEROG	5252.0	0.254570	0.846047	0.000000	0.000000	0.000000	0.000000	10.000000
DELINQ	5380.0	0.449442	1.127266	0.000000	0.000000	0.000000	0.000000	15.000000
CLAGE	5652.0	179.766275	85.810092	0.000000	115.116702	173.466667	231.562278	1168.233561
NINQ	5450.0	1.186055	1.728675	0.000000	0.000000	1.000000	2.000000	17.000000
CLNO	5738.0	21.296096	10.138933	0.000000	15.000000	20.000000	26.000000	71.000000
DEBTINC	4693.0	33.779915	8.601746	0.524499	29.140031	34.818262	39.003141	203.312149

FEATURE IMPORTANCE – TUNED RANDOM FOREST

